Supplementary Materials: Simulation Parameter Tables

CancerEvo: Evolutionary Dynamics of Solid and Liquid Tumors

Theoretical Biology & Mathematical Oncology Framework

June 16, 2026

1 Introduction

This supplementary document summarizes the key parameter values and configurations used in the CancerEvo simulation framework. The model describes the evolutionary dynamics of cancer clones under various spatial and ecological constraints, comparing a local, neighborhood-restricted **Solid Tumor** model with a global, well-mixed **Liquid Tumor** model.

Section 2 details the shared genomic architecture and baseline rates. Section 3 provides the parameters for the ensemble simulations (including diploid, aneuploid, and polyploid missegregation configurations), stability sweeps, parameter phase diagrams, and therapeutic interventions.

2 Genomic Architecture and Baseline Rates

Table 1 outlines the shared genomic architecture and rate parameters that are identical across both the solid and liquid tumor models to ensure direct comparability.

Table 1: Shared Genomic Architecture and Baseline Simulation Rates

Parameter	Symbol	Default Value	Biological / Model Description
Genomic Architecture			
Ploidy Copy Number	N_{CHR}	2[a]	Initial number of chromosome copies per ce
Instability Genes	N_I	10	Loci driving mutation rate increments
Oncogenes	N_O	10	Loci driving division rate increments
Suppressor Genes	N_S	10	Loci driving division rate increments
Missegregation Genes	N_M	5	Loci driving chromosome missegregation
Housekeeping Genes	N_{HK}	10	Essential loci (homozygous mutation is leth
Total Loci per Chromosome	N_{genes}	45	Sum of all gene classes ($N_I + N_O + N_S + N_M +$
Initial Cancer Seed State	–	–	1 mutated I gene (bit 0), 1 mutated O gene (bi
Baseline Rates			
Baseline Division Rate	r_0	0.15	Birth rate of healthy wild-type cells ($r_{\mathrm{WT}} =$
Driver Rate Increment	dr	0.008	Birth rate increase per mutated O or S alle
Maximum Division Rate	$r_{\max}$	0.30	Upper cap on division rate ($2 \times r_0$)
Baseline Mutation Rate	μ_0	0.0	Background mutation rate per locus per divis

[a] Set to $N_{\mathrm{CHR}} = 1$ (haploid) specifically during the stability sweeps.
[b] Represented as chromosome bitmask where all other 43 loci start unmutated.

3 Simulation Configurations and Sweep Parameters

Table 2 details the distinct parameter sets and tissue environments used for the four primary studies in both the Solid and Liquid cases: Ensemble Simulations, Stability Sweeps, Parameter Phase Diagrams, and Therapeutic Interventions.

Table 2: Systematic Overview of Simulation Configuration Parameters

Configuration Variable	Ensemble Simulations	Stability Sweeps	Parameter Phase Diagrams	Therapeutic Interventions
Solid Tumor Model (Local Moran-like competition in 2D Moore neighborhood)				
Lattice Side (L)	200	200	80	200
Total Cells ($N = L^2$)	40,000	40,000	6,400	40,000
Max Simulation Steps (n_{steps})	2500	500	1000	Up to 3000[a]
Ploidy (N_{CHR})	2	1 (Haploid)	2	2
Seeding Strategy	Circular seed ($R = 0.005L$)	Circular seed ($R \approx 0.252L$)	Circular seed ($R = 0.05L$)	Circular seed ($R = 0.05L$)
Initial Tumor Fraction ($f_C(0)$)	$\sim 0.01\%$ (~ 5 cells)	20.0% ($\sim 8,000$ cells)	$\sim 0.8\%$ (~ 49 cells)	$\sim 0.8\%$ (~ 314 cells)
Mutation Rate Increment ($d\mu$)	0.012	Swept $[10^{-6}, 0.02]$[b]	Swept [0.0001, 0.03]	0.012 (0.024 in D)
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	0.0
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	Whole
Replicate Count	500 per ploidy	1 per $r_{\max}$ step	50 per grid point (20×20)	1 per intervention
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.76$ or $f_{\text{WT}} > 0.84$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$
Liquid Tumor Model (Global Moran-like competition with random placement)				
Lattice Side (L)	200	200	80	–[c]
Total Cells ($N = L^2$)	40,000	40,000	6,400	–
Max Simulation Steps (n_{steps})	2500	3000	3000	–
Ploidy (N_{CHR})	2	1 (Haploid)	2	–
Seeding Strategy	Scattered random sites	Scattered random sites	Scattered random sites	–
Initial Seed Count (n_{seed})	10 cells	10 cells	50 cells	–
Mutation Rate Increment ($d\mu$)	0.023 (D) / 0.045 (A, P)	Swept $[10^{-6}, 0.04]$[b]	Swept [0.0001, 0.03]	–
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	–
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	–
Replicate Count	500 per ploidy	1 per $r_{\max}$ step	50 per grid point (20×20)	–
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_C = 1.0$ or $f_C = 0.0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	–

[a] Pre-intervention phase runs up to 2000 steps until the tumor density $f_C \geq 0.30$ trigger is met; the post-intervention phase runs for exactly 1000 steps.
[b] Refined dynamically using a bisection algorithm with 14 iterations for each division rate step.
[c] Therapeutic interventions are implemented and analyzed exclusively for the solid tumor model.
Note on Ensemble Labels: **D** = Diploid ($N_{\text{CHR}} = 2, dm = 0.0$), **A** = Aneuploid ($N_{\text{CHR}} = 2, dm = 0.01$, chunk-based missegregation), **P** = Polyploid ($N_{\text{CHR}} = 2, dm = 0.01$, whole-chromosome missegregation).

4 Therapeutic Intervention Mechanisms (Solid Tumor)

Table 3 describes the targeting rules, execution parameters, and physiological analogues of the four distinct therapeutic interventions simulated in the solid tumor model.

Table 3: Mechanistic Parameters of simulated Therapeutic Interventions

Type	Biological Concept	Model Parameter	Clearance / Mutation Update Rule
A	Fitness-selective targeted therapy	r_i (cell division rate)	Active cancer cells are cleared (reset to WT) with cell-specific probability $P(\text{clear}) = 1 - r_0/r_i$.
B	Driver-selective early clone therapy	$i_{\text{lim}} \leq 1$ ($\mu \leq d\mu$)	Cells belonging to mutation class 0 or 1 are cleared with probability $p_d = 1.0$.
C	Instability-selective late clone therapy	$i_{\text{lim}} \geq 3$ ($\mu \geq 3d\mu$)	Cells belonging to mutation class ≥ 3 are cleared with probability $p_d = 1.0$.
D	Mutagenic load therapy	new $d\mu = 0.024$	The mutation rate increment parameter $d\mu$ is doubled from 0.012 to 0.024, immediately doubling the mutation rates of all living cells.

Common Parameters: Interventions are triggered exactly when the active cancer density $f_C \geq 0.30$ (30% of the tissue area). Displaced or cleared cancer cells are reset to the healthy wild-type state ($r_{\text{cell}} = r_0$, $\mu_{\text{cell}} = \mu_0$, $N_{\text{CHR}} = 2$, chromosomes unmutated).